

Apple Inc. (AAPL) Stock Forecasting and Profitability Analysis

Next-Day Forecasting, Model Comparison and Trading Strategy Back-test

Project Report

Domain: Stock Market Analysis, Prediction and Profitability

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This report is an academic back-test on historical data. Nothing in it constitutes financial or investment advice.

Executive Summary

This project investigated whether the next-day closing price of Apple Inc. (NASDAQ: AAPL) can be forecast from historical price and volume data, and whether such forecasts can be turned into a profitable trading strategy. Six forecasting approaches (Ridge regression, Random Forest, XGBoost, Support Vector Regression, ARIMA and an equal-weight Ensemble of Random Forest, XGBoost and ARIMA) were compared against a naive random-walk benchmark. Their forecasts were then used to drive a long-only trading strategy, which was back-tested against Buy & Hold and a Simple Moving Average (SMA) 50/200 crossover strategy.

The main findings are as follows.

- **Forecast accuracy.** All models achieved a price-level R^2 of approximately 0.99 and a Mean Absolute Percentage Error (MAPE) of about 1.1% on the 2023–2026 test period. The naive random walk achieved the same, so these figures do not demonstrate predictive skill. On daily returns, R^2 was close to zero for every model and no model exceeded the proportion of up-days (54.6%) in directional accuracy.
- **Profitability.** The Ensemble strategy turned \$100,000 into \$231,100 (+131.1%) after transaction costs and was profitable in every calendar year of the test period. Random Forest produced the highest return among the machine-learning strategies (+147.7%). However, Buy & Hold produced the highest return overall (+165.5%, final value \$265,505).
- **Risk.** The Ensemble strategy had a maximum drawdown of 31.9% compared with 33.4% for Buy & Hold, a modest improvement. Several other strategies showed shallower drawdowns, with SVR the lowest at 25.1%, but all at the cost of lower returns.
- **Forward forecast.** Using information available to 11 September 2026, the Ensemble projects a base-case gain of 2.28% for October 2026 (about \$2,279 on a \$100,000 portfolio), with AAPL rising from approximately \$337 to \$345. This is a model estimate with significant uncertainty and should not be interpreted as a reliable prediction.

Overall, the results are consistent with the weak-form efficient market hypothesis: next-day AAPL returns are almost unpredictable from past prices and volume, and the learned models offer at best a marginal improvement over a random walk.

1. Introduction and Objectives

Forecasting equity prices is a long-standing problem in quantitative finance. The weak form of the efficient market hypothesis states that past prices and volumes contain no exploitable information about future prices; if true, no model built on such data can consistently outperform a naive forecast. This project tests that proposition empirically on Apple Inc. using more than two decades of daily data, modern machine-learning methods and a realistic, cost-aware trading simulation.

The objectives of the project were to:

- forecast the next trading day's closing price of AAPL using a set of engineered technical features;
- compare several forecasting models against a naive random-walk benchmark using standard error metrics (MAE, MSE, RMSE, MAPE, R^2) and directional accuracy;
- convert the forecasts into trading signals and measure the profitability of the resulting strategies on an untouched test period, including transaction costs;

- compare the strategies with Buy & Hold and a rule-based technical strategy (SMA 50/200 crossover);
- report daily, weekly, monthly and yearly profit and loss for the main model; and
- produce a week-by-week forward forecast for October 2026.

2. Data

Daily historical data for AAPL was obtained from Yahoo Finance using the yfinance Python library. Prices are adjusted for splits and dividends. The data was cross-checked against an independent second source in a separate validation notebook, and an exploratory data analysis was carried out in a further notebook before modelling.

Table 1. Dataset summary

Item	Value
Instrument	Apple Inc. (NASDAQ: AAPL)
Frequency	Daily (trading days)
Fields	Date, Open, High, Low, Close, Volume, Dividends, Stock Splits
Period covered	22 September 1986 to 11 September 2026
Records	10,070
Missing values / duplicate dates	0 / 0
Modelling rows (after feature construction)	10,020

3. Methodology

3.1 Problem formulation

The prediction target is the next trading day's log return, $r(t+1) = \ln(C(t+1) / C(t))$, where C denotes the closing price. The forecast return is converted back into a closing-price forecast as $C^{\wedge}(t+1) = C(t) \times \exp(r^{\wedge}(t+1))$, so that price-based error metrics can be reported. The forecasting horizon is one trading day.

Returns were modelled rather than raw prices for three reasons. First, the price series is non-stationary, whereas the Augmented Dickey-Fuller test on daily log returns from 2000 onward gives a statistic of -17.25 ($p \approx 6 \times 10^{-30}$), confirming stationarity. Second, the price spans from about \$0.10 to over \$330, and tree-based models cannot extrapolate beyond the price range seen in training. Third, a price-level regression mostly learns that tomorrow's price is approximately today's price.

3.2 Feature engineering

Twenty-three features were constructed using only information available up to and including day t , with look-back windows of 1 to 50 days. Every price-level quantity was normalised by the current close so that features are scale-free and comparable across decades.

Table 2. Feature groups

Group	Features
Returns	Ret_1, Ret_5, Ret_10, Ret_20 (log returns over 1, 5, 10 and 20 days)
Lagged returns	Ret_Lag_1, 2, 3, 5, 10
Trend	Price gap to the 5, 10, 20 and 50-day simple moving averages
Momentum	MACD (12, 26, 9): MACD line, signal line and difference; RSI (14, Wilder smoothing)
Volatility and range	10-day and 20-day rolling volatility of returns; (High – Low) / Close; (Close – Open) / Open
Volume	Daily log change in volume; volume relative to its 20-day average

3.3 Chronological data split

The data was split strictly in time with no shuffling. Data before 2000 was excluded from training because prices below \$1 were quoted in coarse tick increments that distort returns, and because the market regime was very different.

Table 3. Train, validation and test periods

Set	Period	Rows	Purpose
Train	3 Jan 2000 – 31 Dec 2019	5,031	Fit models
Validation	2 Jan 2020 – 30 Dec 2022	756	Compare models; tune trading threshold
Test	3 Jan 2023 – 10 Sep 2026	925	Final out-of-sample evaluation and back-test

To mimic real-world use, every model was refitted before each test year on all data up to 31 December of the previous year (yearly walk-forward retraining). The models were also deliberately regularised (shallow trees, a minimum leaf size of 50, a ridge penalty and a small learning rate) to limit overfitting to a very noisy target.

3.4 Models

Table 4. Forecasting models and rationale

Model	Configuration and rationale
Naive (Random Walk)	Predicted return of zero, so tomorrow's close equals today's close. The benchmark every forecaster must beat.
ARIMA(1,0,1)	Classical time-series model on log returns. $d = 0$ because returns are stationary. Parameters estimated on training data and applied through the test data without refitting within the year.
Ridge regression	Linear baseline over all features with L2 penalty ($\alpha = 10$) to handle correlated indicators.
Random Forest	400 trees, maximum depth 6, minimum leaf size 50. Non-linear and robust to noise; bagging reduces variance.
XGBoost	400 trees, depth 3, learning rate 0.02, subsample 0.8, column sample 0.8, min child weight 20, lambda 5. Gradient boosting with built-in regularisation.
SVR (RBF kernel)	$C = 0.1$, $\epsilon = 0.01$. Trained on the most recent 2,500 rows because of quadratic training cost.
Ensemble	Equal-weight mean of the Random Forest, XGBoost and ARIMA return forecasts. Averaging weakly correlated forecasters usually lowers variance.

Deep-learning models (LSTM or GRU) were considered but not used. Published evidence and the validation results both indicate that next-day returns carry very little exploitable signal, so regularised models whose behaviour can be explained were preferred.

3.5 Evaluation metrics

Price-level accuracy was measured with Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE) and R^2 . Because price-level R^2 is inflated by the persistence of prices, R^2 on returns and directional accuracy (the share of days on which the sign of the forecast return matched the sign of the actual return) were also reported.

3.6 Trading strategy and back-test design

The forecasts were converted into a long-only, all-in/all-out strategy starting with \$100,000, with no short selling and no leverage. A transaction cost of 0.1% was charged on every buy and every sell. Signals are computed after the close of day t and trades are executed at that day's close, an idealised assumption that ignores slippage.

Table 5. Trading rules

Action	Condition
BUY	Not currently holding and forecast return $> +\tau$
SELL	Currently holding and forecast return $< -\tau$
HOLD	Otherwise; keep the current position (cash or shares)

The threshold τ creates a no-trade band that filters out weak signals. It was selected separately for each model on the validation set from the candidates 0%, 0.05%, 0.1%, 0.2% and 0.3%, then frozen for the test period. The selected values were 0.30% for Ridge and SVR, 0.10% for XGBoost, 0.05% for Random Forest, and 0.00% for ARIMA and the Ensemble.

Two baselines were used. Buy & Hold buys on the first test day and holds to the end. The SMA 50/200 crossover is a classic rule-based technical strategy that holds the stock when the 50-day average is above the 200-day average and sells otherwise, with no machine learning.

4. Results

4.1 Validation results (2020–2022)

On the validation set, no model materially improved on the random walk in price terms: MAE ranged from \$2.02 (ARIMA) to \$2.16 (SVR), against \$2.03 for the naive forecast, and MAPE was approximately 1.67% for the leading models. ARIMA delivered the highest directional accuracy (52.9%), followed by Ridge (52.0%) and the Ensemble (51.6%), compared with 51.3% of days being up-days. Margins of this size are small relative to the sampling noise in about 750 observations.

4.2 Test-set forecast accuracy (2023–2026)

Table 6. Forecast metrics on the test set (925 trading days)

Model	MAE (\$)	RMSE (\$)	MAPE (%)	R ² (price)	R ² (return)	Direction acc. (%)
Naive (Random Walk)	2.46	3.63	1.13	0.9939	-0.0044	n/a
Ridge	2.51	3.68	1.15	0.9937	-0.0243	50.4
RandomForest	2.46	3.64	1.13	0.9938	-0.0089	53.4
XGBoost	2.47	3.64	1.13	0.9938	0.0003	53.4
SVR	2.65	3.86	1.21	0.9931	-0.1173	52.1
ARIMA(1,0,1)	2.46	3.64	1.13	0.9938	-0.0060	53.3
Ensemble	2.45	3.63	1.12	0.9939	0.0024	53.7

Key observations:

- **Price-level metrics do not separate the models.** Every model, including the naive benchmark, achieved a price-level R² of about 0.99 and a MAPE of about 1.1%. Today's price already explains almost all of tomorrow's price, so these figures are not evidence of skill.
- **The Ensemble was marginally the best.** It recorded the lowest MAE (\$2.455) and RMSE (\$3.628), compared with \$2.462 and \$3.631 for the naive forecast. The improvement in RMSE is about \$0.004, which is negligible in practice.
- **Return-space R² is approximately zero.** It ranged from slightly negative to marginally positive (0.0024 for the Ensemble and 0.0003 for XGBoost), and SVR (−0.117) and Ridge (−0.024) were worse than predicting zero.
- **Directional accuracy did not beat the base rate.** The Ensemble reached 53.7%, with Random Forest and XGBoost at 53.4%. However, 54.6% of test days were up-days, so a strategy of always predicting an increase would have been more accurate than any model.

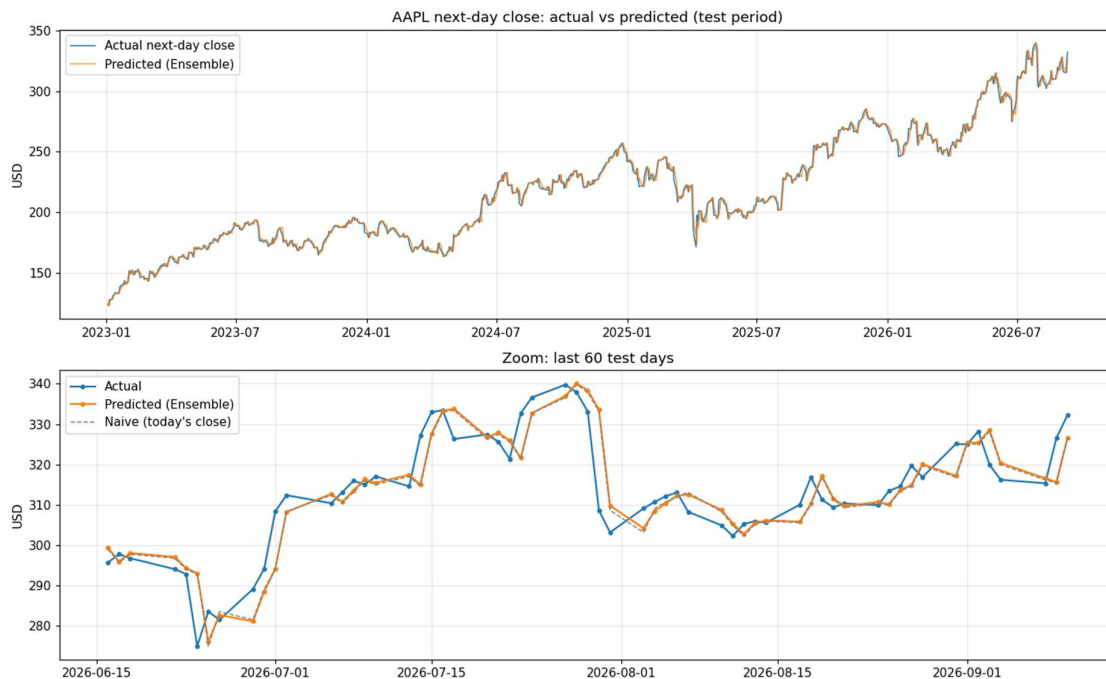


Figure 1. Actual versus predicted next-day close (Ensemble) over the test period, with a zoom on the last 60 days. The predicted series closely tracks the naive forecast (today's close).

4.3 Feature importance

Feature importances from the Random Forest and XGBoost models trained on 2000–2022 are shown in Figure 2. The Random Forest relied most heavily on short-term volatility (Vol_10), the intraday open-to-close range (OC_Range), the 5-day moving-average gap and the 3-day lagged return, followed by the high-low range and the 10-day moving-average gap. XGBoost spread its importance more evenly across features, which is typical when no single predictor carries strong signal. The prominence of volatility and range features suggests that the models capture the size of recent moves more than their direction.

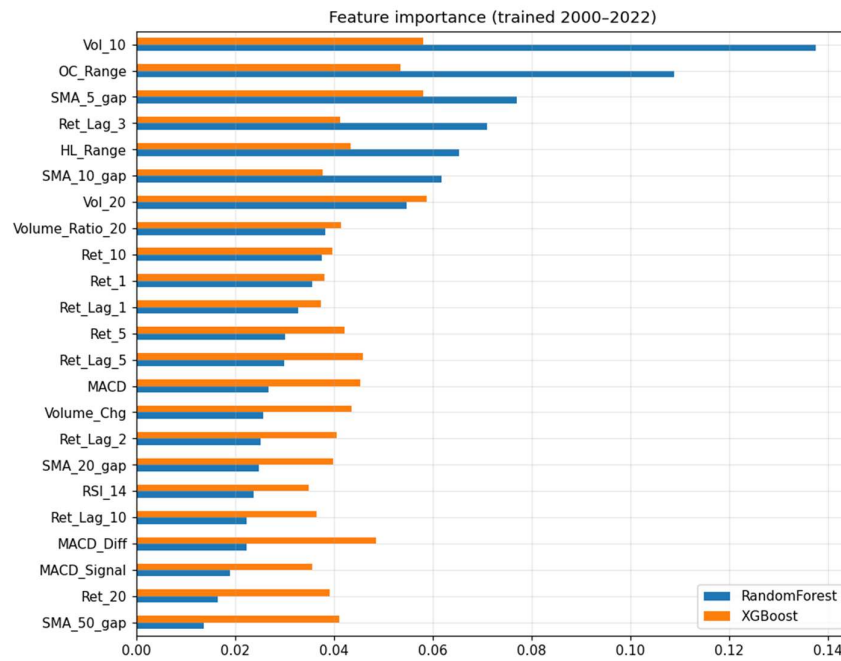


Figure 2. Feature importance of the Random Forest and XGBoost models (trained 2000–2022).

4.4 Back-test results

Table 7. Back-test performance on the test set (initial capital \$100,000; 0.1% cost per trade)

Strategy	τ (%)	Final (\$)	Return (%)	Ann. return (%)	Sharpe	Max DD (%)	Trades	Win rate (%)
Buy & Hold	-	265,505	165.5	30.5	1.16	-33.4	1	-
SMA 50/200 crossover	-	129,037	29.0	7.2	0.44	-30.0	5	50.0
Ridge	0.30	206,347	106.3	21.8	0.95	-28.4	23	54.5
RandomForest	0.05	247,705	147.7	28.0	1.11	-31.9	19	77.8
XGBoost	0.10	215,860	115.9	23.3	0.97	-28.6	35	52.9
SVR	0.30	185,218	85.2	18.3	0.87	-25.1	179	60.7
ARIMA(1,0,1)	0.00	180,910	80.9	17.5	0.77	-36.1	88	68.2
Ensemble	0.00	231,100	131.1	25.6	1.03	-31.9	39	68.4

Key observations:

- **Buy & Hold was the strongest strategy.** It returned 165.5% (annualised 30.5%, Sharpe ratio 1.16), with a final value of \$265,505. AAPL rose strongly during the test period, and strategies that spent time in cash missed part of the advance.

- **Random Forest was the best machine-learning strategy.** It returned 147.7% (annualised 28.0%, Sharpe 1.11) with only 19 trades and a 77.8% win rate over closed round trips. It was followed by the Ensemble (131.1%), XGBoost (115.9%) and Ridge (106.3%).
- **All learned strategies were profitable and all beat the SMA crossover.** The SMA 50/200 crossover returned only 29.0% (Sharpe 0.44), spending long periods out of the market, notably during 2025.
- **Risk reduction was limited.** Maximum drawdowns of the model-based strategies ranged from 25.1% (SVR) to 36.1% (ARIMA) against 33.4% for Buy & Hold. The Ensemble and Random Forest both had a drawdown of 31.9%.
- **Rankings were not stable between validation and test.** ARIMA was the best strategy on validation (+138.3%) but one of the weakest on the test set (+80.9%, the worst drawdown). Random Forest rose from second on validation to first among the learned strategies on test. This instability is typical of noisy daily signals and cautions against selecting a model on the basis of a single period.

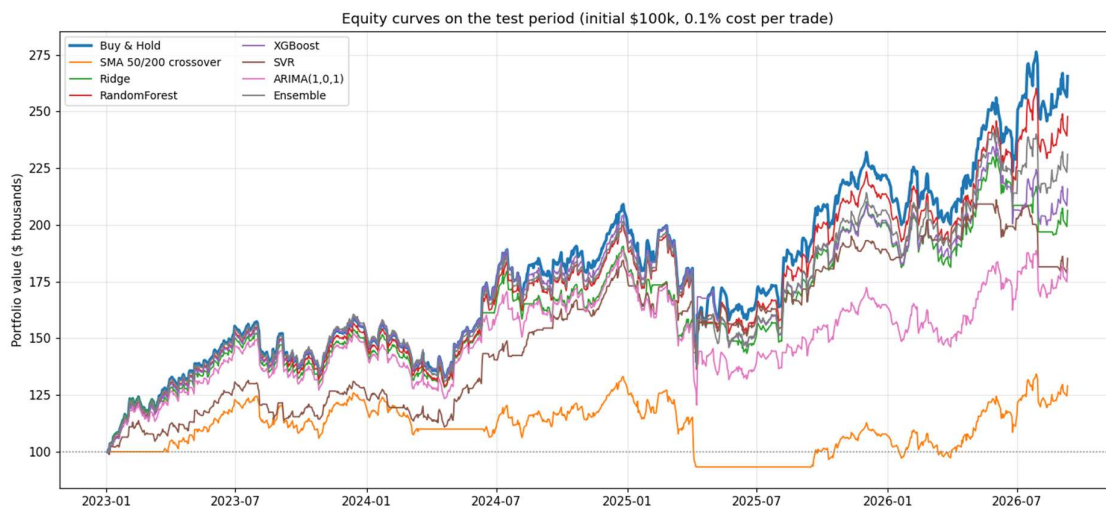


Figure 3. Equity curves of all strategies over the test period (initial \$100,000, 0.1% cost per trade).

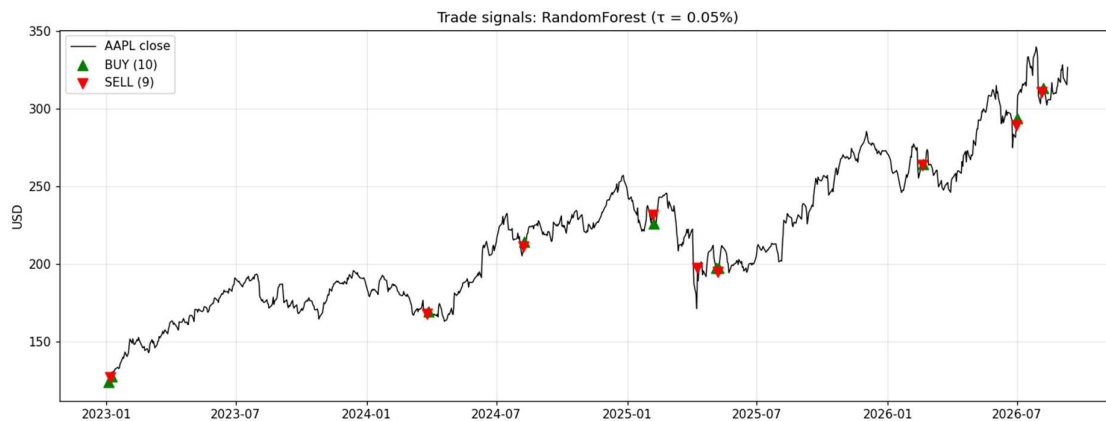


Figure 4. Buy and sell signals generated by the Random Forest strategy ($\tau = 0.05\%$) over the test period.

4.5 Sensitivity to transaction costs

To test robustness, the test-set back-test was repeated under transaction costs of 0%, 0.10%, 0.25% and 0.50% per trade.

Table 8. Final portfolio value (\$) under different transaction costs

Strategy	0.00%	0.10%	0.25%	0.50%
Buy & Hold	265,770	265,505	265,106	264,442
Ridge	211,150	206,347	199,337	188,158
Random Forest	252,459	247,705	240,733	229,525
XGBoost	223,552	215,860	204,801	187,580
SVR	221,544	185,218	141,537	90,322
ARIMA(1,0,1)	197,560	180,910	158,502	127,096
Ensemble	240,296	231,100	217,946	197,627

Buy & Hold is almost unaffected by costs because it trades once. Strategies that trade frequently are highly cost-sensitive: SVR (179 trades) loses more than half of its final value between the 0% and 0.50% cases and ends below its starting capital at 0.50%, while ARIMA (88 trades) loses over a third. Random Forest and the Ensemble, which trade far less often, are more resilient. No learned strategy outperformed Buy & Hold at any cost level.

4.6 Profitability of the main model (Ensemble)

The Ensemble was designated the main model because averaging several forecasters gives more stable forecasts than relying on any one, and because it recorded the lowest test-set RMSE. Starting with \$100,000, it ended the test period at \$231,100, a profit of \$131,100. The tables and figure below break this result down by period.

Table 9. Ensemble yearly profit and loss compared with Buy & Hold

Year	Start (\$)	End (\$)	Profit/Loss (\$)	Return (%)	Buy & Hold return (%)
2023	100,000	156,061	56,061	56.06	54.64
2024	156,061	194,860	38,799	24.86	30.71
2025	194,860	205,090	10,230	5.25	9.05
2026 (to 10 Sep)	205,090	231,100	26,010	12.68	20.45

The Ensemble was profitable in all four calendar years. It marginally beat Buy & Hold in 2023 (56.1% against 54.6%) but trailed it in 2024, 2025 and 2026, consistent with the strategy sitting out part of a rising market.

Table 10. Frequency of profitable periods (Ensemble)

Period	Count	Profitable	Loss-making	% profitable	Average P/L (\$)	Best (\$)	Worst (\$)
Daily	925	484	426	52.3	141.73	20,893.75	-17,307.31
Weekly	193	115	77	59.6	679.27	20,193.69	-23,513.35
Monthly	45	29	16	64.4	2,913.33	31,547.72	-26,325.66
Yearly	4	4	0	100.0	32,774.99	56,061.12	10,229.78

The share of profitable periods rises with the length of the period, from 52.3% of days to 59.6% of weeks and 64.4% of months. Losses are a normal feature of the strategy: the worst single day lost

\$17,307, the worst week \$23,513 and the worst month \$26,326 (June 2026, -10.95%). Note that some days have zero profit and loss because the strategy was in cash, so profitable and loss-making counts do not sum to the number of days.

Table 11. Ensemble monthly profit and loss, last 12 months

Month	Start (\$)	End (\$)	Profit/Loss (\$)	Return (%)
2025-10	189,328	202,162	12,834	6.78
2025-11	202,162	208,705	6,543	3.24
2025-12	208,705	205,090	-3,615	-1.73
2026-01	205,090	195,751	-9,339	-4.55
2026-02	195,751	201,964	6,214	3.17
2026-03	201,964	195,292	-6,672	-3.30
2026-04	195,292	208,805	13,513	6.92
2026-05	208,805	240,352	31,548	15.11
2026-06	240,352	214,027	-26,326	-10.95
2026-07	214,027	218,043	4,016	1.88
2026-08	218,043	224,222	6,179	2.83
2026-09	224,222	231,100	6,878	3.07



Figure 5. Ensemble strategy: yearly return compared with Buy & Hold, and monthly and weekly profit and loss.

4.7 October 2026 forward forecast

A week-by-week forecast for October 2026 was generated with the same Ensemble methodology. The Random Forest, XGBoost and ARIMA models were refitted on all labelled observations through 11 September 2026, the last date in the dataset. Because the dataset ends on that date, this is a model-based forecast made as of 11 September 2026 and does not use later information.

Future open, high, low and volume values are not available, so they were generated under simple assumptions in order to compute features recursively: the next-day open equals the previous close, the high-low range equals the 20-day median range (2.08% of price), and volume equals the 20-day median (39,128,150 shares). Predicted closes were fed back into the feature calculation day by day, and ARIMA forecasts were generated independently. The daily Ensemble return was the mean of the three model forecasts. Since the Ensemble threshold τ is 0%, a positive forecast return produces a BUY/HOLD LONG signal.

Table 12. October 2026 weekly forecast (Ensemble, \$100,000 starting capital)

Week (2026)	Trading days	Start price (\$)	End price (\$)	Predicted return (%)	Predicted profit (\$)	Projected portfolio (\$)
Oct 01 - Oct 02	2	337.29	337.98	0.21	205.08	100,205.08
Oct 05 - Oct 09	5	337.98	339.75	0.52	525.20	100,730.27
Oct 12 - Oct 16	5	339.75	341.49	0.51	514.45	101,244.72
Oct 19 - Oct 23	5	341.49	343.23	0.51	515.88	101,760.60
Oct 26 - Oct 30	5	343.23	344.98	0.51	518.15	102,278.75

The forecast signals BUY/HOLD LONG on every day of October. The predicted price rises from about \$337.29 at the start of the month to \$344.98 by 30 October, a base-case return of 2.28%, equivalent to a profit of \$2,278.75 and an ending portfolio of \$102,278.75.

This forecast should be treated with considerable caution. The forecast daily return is nearly constant at about +0.1% per day, which is far more persistent than real daily returns. This is largely a consequence of the recursive method: the ARIMA component contributes a near-constant positive drift of about 0.09% per day, and the tree models, fed synthetic inputs built from their own predictions, produce smooth, slightly positive forecasts. The month-on-month volatility actually observed in AAPL, shown by the historical monthly profit and loss in Section 4.6, is many times larger than the forecast gain. The projection is therefore best read as the model's central tendency rather than a prediction of the likely path.

5. Discussion

The results tell a consistent story. Price-level accuracy metrics are almost identical across all models and the naive benchmark, because daily price changes are small relative to the price level. Once attention moves to returns, the models explain essentially none of the variation (R^2 close to zero), and their ability to call the direction of the next move is no better than assuming that the stock always rises, which was true on 54.6% of test days. This matches the weak-form efficient market hypothesis and the published literature on short-horizon return prediction.

The profitability analysis shows why accuracy and profit are different questions. Every learned strategy made money in the test period, but this largely reflects AAPL's strong upward trend, which any long-exposed strategy participates in. Because the strategies exit the market on weak or negative signals, they forgo some of that trend, and none of them outperformed simply holding the stock. The benefit of

timing, a reduction in downside risk, was modest here: the Ensemble reduced the maximum drawdown by only 1.5 percentage points relative to Buy & Hold.

The instability of model rankings between validation and test (ARIMA first on validation, weakest drawdown on test) and the strong sensitivity of frequent traders to transaction costs both reinforce the conclusion that any apparent edge in daily signals is fragile. The higher win rates reported for some strategies (for example 77.8% for Random Forest) are measured over small numbers of closed trades (for Random Forest, only nine round trips) and should be interpreted with caution.

Several design choices strengthened the reliability of the findings: all evaluation is strictly out-of-sample and forward in time, thresholds were selected using validation data only, models were retrained annually in a walk-forward manner, features are scale-free, and transaction costs and cost sensitivity were included.

6. Conclusions

Forecasting. Every model reached a price-level R^2 of about 0.99 and a MAPE of about 1.1%, but so did the naive random walk, so these numbers do not indicate skill. On returns, R^2 is approximately zero and directional accuracy (at best 53.7%) is below the 54.6% share of up-days. The learned models, with the Ensemble marginally ahead, give at most a negligible improvement over the random walk.

Profitability. The main model (Ensemble) is profitable overall (+131.1%, \$100,000 to \$231,100) and in every year of the test period, and it beats the rule-based SMA 50/200 strategy (+29.0%). However, Buy & Hold delivered the highest total return (+165.5%) and the best risk-adjusted return (Sharpe 1.16). The benefit of the model-based strategies is limited to a modest reduction in drawdown for some of them.

Robustness. Results are out-of-sample, use validation-selected thresholds, walk-forward retraining and transaction costs. Model rankings changed between validation and test, and frequent-trading strategies were highly sensitive to cost assumptions, which is typical of weak daily signals.

Forward view. The Ensemble's base-case October 2026 forecast is a gain of 2.28% (AAPL from about \$337 to \$345; about \$2,279 on \$100,000). Given the recursive construction and the low predictability of daily returns, it should be regarded as indicative only.

Overall. The project supports the view that next-day AAPL returns are very difficult to predict from historical prices and volume alone, and that a simple, low-turnover exposure to the stock was at least as effective as the machine-learning timing strategies tested.

7. Limitations and Future Work

- Execution is idealised: trades occur at the same-day close with no slippage, bid-ask spread, taxes or market impact.
- Only one stock was analysed, over one market regime that was mostly favourable to long positions. Results may not generalise to other securities or to prolonged bear markets.
- Only technical (price and volume) features were used. Fundamental data, news and sentiment were not included.
- The October 2026 forecast relies on synthetic future OHLCV values and recursive prediction, which compounds model error and understates real uncertainty. No confidence interval was produced.

- Strategy parameters (τ) were chosen from a small grid on a single validation window, and the number of closed trades for several strategies is small, limiting statistical reliability. No formal significance testing of performance differences was carried out.

Possible extensions include adding sentiment and macroeconomic features, modelling weekly or monthly horizons directly, classifying up and down moves and sizing positions by predicted probability, introducing short selling and risk controls such as stop-losses, evaluating several stocks, applying formal significance tests (for example the Diebold-Mariano test), and testing sequence models such as LSTM or GRU.

Appendix: Reproducibility

All results in this report were produced by a single Python notebook (Apple_Forecast.ipynb) using pandas, NumPy, scikit-learn, XGBoost and statsmodels, with a fixed random seed of 42. Supporting output files include the test-set forecast metrics, back-test results, Ensemble profit tables at daily, weekly, monthly and yearly frequency, the October 2026 daily and weekly forecasts, and the figures reproduced in this report.